Proof. Since σ is injective,

$$h_i^{(S)}(\boldsymbol{z}) = h_i^{(S')}(\boldsymbol{z}) \qquad \forall \boldsymbol{z} \in \mathbb{R}^d.$$

Hence

$$\left(\boldsymbol{w}_i^{(S)} - \boldsymbol{w}_i^{(S')}\right)^\intercal \boldsymbol{z} + \left(b_i^{(S)} - b_i^{(S')}\right) = 0 \qquad \forall \boldsymbol{z} \in \mathbb{R}^d.$$

Taking $\boldsymbol{z} = \mathbf{0}$ gives $b_i^{(S)} = b_i^{(S')}$, and then

$$\left(\boldsymbol{w}_i^{(S)} - \boldsymbol{w}_i^{(S')}\right)^\intercal \boldsymbol{z} = 0 \qquad \forall \boldsymbol{z} \in \mathbb{R}^d,$$

implies $\boldsymbol{w}_i^{(S)} = \boldsymbol{w}_i^{(S')}$, since the equation holds for any point in $\mathbb{R}^d$. □

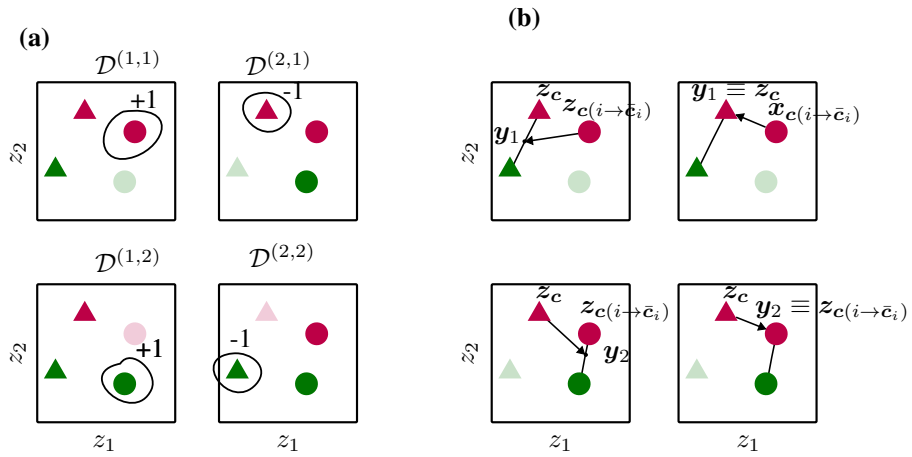

Figure 13. **Illustration of the invariance lemma (left) and the main proposition (right). (a)** The invariance lemma (Lemma 3): any point $\boldsymbol{z_c}$ can be made a support vector by an appropriate choice of a pair of datasets. **(b)** The main proposition (Proposition 1): linearity and orthogonality under GD+BCE follow because, for each concept i, the SVM support vectors correspond to counterfactual pairs that differ only in the i-th concept. Concretely, when the minority support point is $\boldsymbol{z}_{\boldsymbol{c}(i\to\bar{c}_i)}$, the majority-class support vector $\boldsymbol{y}_1$ under $\mathcal{D}^{(1,1)}$ reduces to $\boldsymbol{z_c}$, and analogously for $\boldsymbol{y}_2$.

Lemma 2 (Bi-directional tight support vectors)**.** For binary concepts $\mathcal{C}_i = \{0, 1\}$ for all $i \in [k]$, consider a pair of cross-datasets $\mathcal{D}^{\boldsymbol{c}}$ and $\mathcal{D}^{\boldsymbol{c}(i\to\bar{c}_i)}$ for some $i \in [k]$ and the corresponding SVM solutions $\{\boldsymbol{w}_i, b_i\}$ and $\{\boldsymbol{w}'_i, b'_i\}$ for these datasets, respectively. Then, $\boldsymbol{z}_{\boldsymbol{c}(i\to\bar{c}_i)}$ is a tight support vector for concept i under $(\boldsymbol{w}_i, b_i)$ trained on $\mathcal{D}^{\boldsymbol{c}}$, and $\boldsymbol{z_c}$ is a tight support vector under $(\boldsymbol{w}'_i, b'_i)$ trained on $\mathcal{D}^{\boldsymbol{c}(i\to\bar{c}_i)}$, i.e. the following hold:

$$\begin{aligned} (\boldsymbol{w}')_i^\intercal \boldsymbol{z_c} + b'_i &= y_i(\boldsymbol{c}) \\ \boldsymbol{w}_i^\intercal \boldsymbol{z}_{\boldsymbol{c}(i\to\bar{c}_i)} + b_i &= y_i(\boldsymbol{c}(i \to \bar{c}_i)), \end{aligned} \tag{8}$$

where $y_i(\boldsymbol{c}) \in \{+1, -1\}$ is the label of $\boldsymbol{z_c}$ with respect to concept i.

Proof. For the second equality: by Remark 1, $N_{i,c_i}(\mathcal{D}^{\boldsymbol{c}}) = 1 + (k-1)(n-1)$ and $N_{i,\bar{c}_i}(\mathcal{D}^{\boldsymbol{c}}) = 1$. Hence both classes are non-empty in $\mathcal{D}^{\boldsymbol{c}}$; standard hard-margin SVM argument then gives at least one support vector in each class achieving equality at the margin (Cortes & Vapnik, 1995). Repeating the same argument for $\mathcal{D}^{\boldsymbol{c}(i\to\bar{c}_i)}$ gives the first equation. □

Lemma 3 (Invariance to irrelevant concepts)**.** Assume each concept is binary, $\mathcal{C}_i = \{0, 1\}$ for all $i \in [k]$, and Desiderata 1–3 hold, and consider either the rule $|T| = 2^{k-1} + 1$ or the cross-dataset design (Definition 6). Then, for any $i \in [k]$ and any $\boldsymbol{c}, \boldsymbol{c}' \in [2]^k$ with $c_i = c'_i$, it holds that

$$P(C_i = c_i \mid \boldsymbol{z_c}) = P(C_i = c_i \mid \boldsymbol{z_{c'}}). \tag{9}$$

Proof. We encode the i-label by $y_i(\boldsymbol{z}) \in \{+1, -1\}$ with $y_i(\boldsymbol{z}) = +1$ iff $C_i(\boldsymbol{z}) = 1$ and -1 otherwise, and let

$$h_i(\boldsymbol{z}) \;:=\; \boldsymbol{w}_i^\intercal \boldsymbol{z} + b_i \tag{10}$$

By Stability (Desideratum 3) and Lemma 1, the affine separator $(\boldsymbol{w}_i, b_i)$ is the same across valid cross-datasets. For any $\boldsymbol{c} \in [2]^k$, consider the cross-dataset $\mathcal{D}^{\boldsymbol{c}(i\to\bar{c}_i)}$. By Remark 1, in this dataset concept i has count 1 for value c_i and count $1 + (k-1)(n-1)$ for value $\bar{c}_i$, so

the unique minority example with respect to concept i is $\boldsymbol{z_c}$. By Lemma 2, both classes have tight support vectors. Since the minority class has exactly one point, that point $\boldsymbol{z_c}$ is the tight support vector for its class. Hence

$$y_i(\boldsymbol{z_c})h_i(\boldsymbol{z_c}) = 1. \tag{11}$$

The same argument applies to $\boldsymbol{c}(i \to \bar{c}_i)$ in $\mathcal{D}$, where it follows that $y_i(\boldsymbol{z}_{\boldsymbol{c}(i\to\bar{c}_i)})\, h_i(\boldsymbol{z}_{\boldsymbol{c}(i\to\bar{c}_i)}) = 1$.

Since this was performed for any $\boldsymbol{c}$, it follows that

$$h_i(\boldsymbol{z}) = \begin{cases} +1, & \text{if } C_i(\boldsymbol{z}) = 1, \\ -1, & \text{if } C_i(\boldsymbol{z}) = 0, \end{cases} \quad \text{on the whole grid } \{\boldsymbol{z_c} : \boldsymbol{c} \in [2]^k\}.$$

Hence $h_i(\boldsymbol{z})$ depends only on $C_i(\boldsymbol{z})$ and not on the other concepts. Since $P(C_i = 1 \mid \boldsymbol{z}) = \sigma(h_i(\boldsymbol{z})) = \frac{1}{1+e^{-h_i(\boldsymbol{z})}}$ (and $P(C_i = 0 \mid \boldsymbol{z}) = 1 - P(C_i = 1 \mid \boldsymbol{z})$), the conditional probability $P(C_i = c_i \mid \boldsymbol{z_c})$ is constant over all $\boldsymbol{c}$ with fixed c_i. In particular, for any $\boldsymbol{c}, \boldsymbol{c}'$ with $c_i = c_i'$,

$$P(C_i = c_i \mid \boldsymbol{z_c}) = P(C_i = c_i' \mid \boldsymbol{z_{c'}}).$$

□

Next, we state an important property of SVMs on two separable sets and adapt it to our case, where one of the elements in the set is a singleton.

Lemma 4 (SVM geometry for separable sets)**.** Given a set of points $\mathcal{Y} := \{\boldsymbol{y}_i\}_{i=1}^N$ with $\boldsymbol{y}_i \in \mathbb{R}^d$, and a point $\boldsymbol{z} \in \mathbb{R}^d$, let $(\boldsymbol{w}, b)$ be the optimal hard-margin SVM separator between classes $\mathcal{Y}$ and $\{\boldsymbol{z}\}$, with the canonical scaling $\boldsymbol{w}^\intercal \boldsymbol{x} + b = \pm 1$ on support vectors. Then:

1. There exist coefficients $\{\lambda_i\}_{i=1}^N$ with $\lambda_i \geq 0$ and $\sum_i \lambda_i = 1$ such that

$$\boldsymbol{w}^\intercal \left(\sum_i \lambda_i \boldsymbol{y}_i \right) + b = -1, \tag{12}$$

$$\boldsymbol{w}^\intercal \boldsymbol{z} + b = +1. \tag{13}$$

2. The weight $\boldsymbol{w}$ is related to the shortest-segment between the convex hulls of the two classes as

$$\frac{2}{\|\boldsymbol{w}\|^2}\boldsymbol{w} = \boldsymbol{z} - \sum_i \lambda_i \boldsymbol{y}_i. \tag{14}$$

Proof. This is a standard geometric characterization of hard-margin SVMs: $\boldsymbol{w}$ is parallel to the shortest segment joining the convex hulls of the two classes, and under canonical margin scaling the two supporting hyperplanes are at signed distance $1/\|\boldsymbol{w}\|$ from the decision hyperplane, hence the support-point displacement equals $\frac{2}{\|\boldsymbol{w}\|^2}\boldsymbol{w}$ (Bennett & Bredensteiner, 2000). □

We now establish the main result on the resulting geometry of linearly generalizable compositional models. Intuition of it, together with the invariance lemma above is presented in Fig. 13.

Proposition 1 (Compositional generalization implies linear factorization)**.** Let $\Pi = (f, \mathcal{H}, A, \mathcal{T})$ be the tuple instantiated in Definition 4, with linear heads $\mathcal{H}$ and A given by GD+CE. Suppose that the training sets follow random sampling with validity rule $R(T) = 1$ if $|T| = 2^{k-1} + 1$. Assume Desiderata 1–3 are satisfied. Then, under the binary grid $\mathcal{C}_i = \{0, 1\}$ with $\mathcal{Z} = \{\boldsymbol{z_c} : \boldsymbol{c} \in [2]^k\} \subset \mathbb{R}^d$, there exist $\{\boldsymbol{u}_{i,0}, \boldsymbol{u}_{i,1} \in \mathbb{R}^d\}_{i=1}^k$ such that for every $\boldsymbol{c} \in [2]^k$ the following holds:

1. *(Linearity)* $\boldsymbol{z_c} = \sum_{i=1}^k \boldsymbol{u}_{i,c_i}$.
2. *(Cross-concept orthogonality)* $(\boldsymbol{u}_{i,1} - \boldsymbol{u}_{i,0}) \perp (\boldsymbol{u}_{j,1} - \boldsymbol{u}_{j,0})$ for all $i, j \in [k]$ with $(i \neq j)$.

Proof. Although Proposition 1 is stated for the $|T| = 2^{k-1} + 1$ regime, the geometric mechanism is easiest to see on the cross-dataset construction $\mathcal{D}^{\boldsymbol{c}}$ (Definition 6), which isolates one-concept flips around a center point $\boldsymbol{c}$ (in the binary case, $|\mathcal{D}^{\boldsymbol{c}}| = 1 + k$). We therefore present the proof using cross-datasets to keep the SVM geometry and the stability constraints transparent.

Linearity.

The idea is to show that for a pair of cross-datasets that share the datapoints in negative class, the shortest distance from a single point in the positive class to the convex set of the positive points is achieved by considering a flip in one of the concepts. We make this concrete below.

Consider any point $\boldsymbol{z_c}$ and its corresponding cross-dataset $\mathcal{D}^{\boldsymbol{c}}$. For any concept $i \in [k]$, let $\boldsymbol{z}_{\boldsymbol{c}(i\to\bar{c}_i)}$ be the counterfactual point obtained by flipping concept i to $\bar{c}_i$, and consider the neighboring cross-dataset $\mathcal{D}^{\boldsymbol{c}(i\to\bar{c}_i)}$.

Note that for the concept i it holds that:

1. Under $\mathcal{D}^{\boldsymbol{c}} = \{\boldsymbol{z}_{\boldsymbol{c}}\} \cup \{\boldsymbol{z}_{\boldsymbol{c}(j \to \bar{c}_j)} : j \in [k]\}$. For each concept i, the marginal counts are

$$N_{i,c_i}(\mathcal{D}^{\boldsymbol{c}}) = k, \qquad N_{i,\bar{c}_i}(\mathcal{D}^{\boldsymbol{c}}) = 1 \tag{15}$$

(by Remark 1). Thus $\boldsymbol{z}_{\boldsymbol{c}(j \to \bar{c}_j)}$ is the unique minority example for concept j (with label $\bar{c}_j$), and

$$\mathcal{Y}_1 := \mathcal{D}^{\boldsymbol{c}} \setminus \{\boldsymbol{z}_{\boldsymbol{c}(i \to \bar{c}_i)}\} \tag{16}$$

is the set of k majority examples (with label c_i).

2. Under

$$\mathcal{D}^{\boldsymbol{c}(i \to \bar{c}_i)} := \{\boldsymbol{z}_{\boldsymbol{c}(i \to \bar{c}_i)}\} \cup \{\boldsymbol{z}_{\boldsymbol{c}}\} \cup \{\boldsymbol{z}_{\boldsymbol{c}(i \to \bar{c}_i,\, \ell \to \bar{c}_\ell)} : \ell \in [k] \setminus \{i\}\},$$

for any $\ell \neq i$ the counts are unchanged: $N_{\ell,c_\ell}(\mathcal{D}^{\boldsymbol{c}(i \to \bar{c}_i)}) = k$ and $N_{\ell,\bar{c}_\ell}(\mathcal{D}^{\boldsymbol{c}(i \to \bar{c}_i)}) = 1$, but for concept i they swap: $N_{i,\bar{c}_i}(\mathcal{D}^{\boldsymbol{c}(i \to \bar{c}_i)}) = k$ and $N_{i,c_i}(\mathcal{D}^{\boldsymbol{c}(i \to \bar{c}_i)}) = 1$. Thus $\boldsymbol{z}_{\boldsymbol{c}}$ is now the unique minority example for concept i (label c_i). We denote

$$\mathcal{Y}_2 = \mathcal{D}^{\boldsymbol{c}(i \to \bar{c}_i)} \setminus \{\boldsymbol{z}_{\boldsymbol{c}}\} \tag{17}$$

as the majority examples for concept i.

Let the pair of majority support vectors for $\mathcal{D}^{\boldsymbol{c}}$ and $\mathcal{D}^{\boldsymbol{c}(i \to \bar{c}_i)}$ be $\boldsymbol{y}_1$ and $\boldsymbol{y}_2$ respectively. By Lemma 4, we can write

$$\boldsymbol{y}_1 = \lambda_i \boldsymbol{z}_{\boldsymbol{c}} + \sum_{j \in [k] \setminus \{i\}} \lambda_j \boldsymbol{z}_{\boldsymbol{c}(j \to \bar{c}_j)} \qquad \text{and} \qquad \boldsymbol{y}_2 = \gamma_i \boldsymbol{z}_{\boldsymbol{c}(i \to \bar{c}_i)} + \sum_{j \in [k] \setminus \{i\}} \gamma_j \boldsymbol{z}_{\boldsymbol{c}(i \to \bar{c}_i,\, j \to \bar{c}_j)} \tag{18}$$

for some convex combinations $\lambda_j \geq 0$ with $\sum_j^k \lambda_j = 1$ and $\gamma_j \geq 0$ with $\sum_j^k \gamma_j = 1$.

Additionally, note that by Lemma 3 it holds for any point $\boldsymbol{z}_{\boldsymbol{c}'}$ for concept $j \in [k]$ that

$$\boldsymbol{w}_j^\intercal \boldsymbol{z}_{\boldsymbol{c}'} + b_j = y_j(\boldsymbol{c}'), \tag{19}$$

where we use a shorthand $y_j(\boldsymbol{c}) = 1$ if $c_j = 1$ and $y_j(\boldsymbol{c}) = -1$ otherwise.

Then, by Lemma 4 it holds that the support vectors are aligned with the shortest segment between the convex sets (pairs of $\boldsymbol{z}_{\boldsymbol{c}(i \to \bar{c}_i)}$ and $\boldsymbol{y}_1$, and $\boldsymbol{z}_{\boldsymbol{c}}$ and $\boldsymbol{y}_2$)

$$\boldsymbol{z}_{\boldsymbol{c}(i \to \bar{c}_i)} + y_i(\boldsymbol{c}) \frac{2}{||\boldsymbol{w}_i||^2} \boldsymbol{w}_i = \boldsymbol{y}_1 \quad \text{ and } \boldsymbol{z}_{\boldsymbol{c}} - y_i(\boldsymbol{c}) \frac{2}{||\boldsymbol{w}_i||^2} \boldsymbol{w}_i = \boldsymbol{y}_2, \tag{20}$$

where clearly $y_i(\boldsymbol{c}(i \to \bar{c}_i)) = -y_i(\boldsymbol{c})$. From this, it follows that

$$\boldsymbol{y}_1 - \boldsymbol{z}_{\boldsymbol{c}(i \to \bar{c}_i)} = \boldsymbol{z}_{\boldsymbol{c}} - \boldsymbol{y}_2. \tag{21}$$

Now, for any $j \neq i$ it follows that

$$\begin{aligned}
\boldsymbol{w}_j^\intercal \boldsymbol{y}_1 + b_j &= \boldsymbol{w}_j^\intercal \left(\lambda_i \boldsymbol{z}_{\boldsymbol{c}} + \sum_{l \in [k] \setminus \{i\}} \lambda_l \boldsymbol{z}_{\boldsymbol{c}(l \to \bar{c}_l)} \right) + b_j && (22)\\
&= \lambda_i \boldsymbol{w}_j^\intercal \boldsymbol{z}_{\boldsymbol{c}} + \sum_{l \in [k] \setminus \{i\}} \lambda_l \boldsymbol{w}_j^\intercal \boldsymbol{z}_{\boldsymbol{c}(l \to \bar{c}_l)} + \sum_{l}^{k} \lambda_l b_j && (23)\\
&= \lambda_i (\boldsymbol{w}_j^\intercal \boldsymbol{z}_{\boldsymbol{c}} + b_j) + \sum_{l \in [k] \setminus \{i\}} \lambda_l (\boldsymbol{w}_j^\intercal \boldsymbol{z}_{\boldsymbol{c}(l \to \bar{c}_l)} + b_j) && (24)\\
&= \lambda_i y_j(\boldsymbol{c}) + \sum_{l \in [k] \setminus \{i,j\}} \lambda_l y_j(\boldsymbol{c}(l \to \bar{c}_l)) + \lambda_j y_j(\boldsymbol{c}(j \to \bar{c}_j)) && (25)\\
&= \lambda_i y_j(\boldsymbol{c}) + \left(\sum_{l \in [k] \setminus \{i,j\}} \lambda_l \right) y_j(\boldsymbol{c}) - \lambda_j y_j(\boldsymbol{c}) && (26)\\
&= (1 - \lambda_j) y_j(\boldsymbol{c}) - \lambda_j y_j(\boldsymbol{c}) = (1 - 2\lambda_j) y_j(\boldsymbol{c}), && (27)
\end{aligned}$$

where we used the fact that λ are convex combinations in the second equality, and the fact that in the paired dataset k-concept values remain the same when flipping any other concept than k.